

All solutions should be well justified. The total amount of points is 90; to pass, you need to get at least 40. Use the departments papers only. Begin the solution of each assignment **on top of a new sheet**.

Do **not** use **red** pen or pencil. You are allowed to use a pocket calculator.

Write your identifier on each sheet.

1. Suppose $\Omega = \{1, 2, 3\}$. Construct all sigma-algebras on Ω which have no more than 4 elements. (10p)
2. The frequency of a certain disease in the general population is $\alpha \in (0, 1)$. There is a fairly accurate medical test for this disease. The test *will* indicate that you have this disease with probability 0.99 in case you really have it; and with probability 0.05 it will indicate that you have the disease when you do not have it. After taking the test, which came out positive (indicating that you have a disease), the doctor told you that now the probability that you have the disease is, in fact, $\frac{1}{11}$. What is the value of α ? (Please give an answer as a rational fraction). (10p)
3. Suppose (X, Y) are jointly continuous with the density $f_{X,Y}(x, y) = 1$ for (x, y) inside of the unit square $0 \leq x \leq 1, 0 \leq y \leq 1$ and $f_{X,Y}(x, y) = 0$ for the other values. Compute the density of random variable $Z = X + Y$. (20p)
4. Let the joint density of random variables X, Y be given by

$$f_{X,Y}(x, y) = c \exp \left\{ -\frac{x^2 + xy + y^2}{2} \right\}$$

- (a) Determine the value of constant c for which $f_{X,Y}$ is, indeed, a density. (10p)
- (b) Compute $E(X^2 \mid Y = y)$. (10p)
5. You toss a coin until you get “head” for the first time. On each toss, the probability of getting “head” is the same and equals Y , where Y itself was originally chosen at random, and has the density $f_Y(y) = 2y$ for $0 < y < 1$ and $= 0$ otherwise. Conditionally on Y , the tosses of the coin are independent. Let $X \in \{1, 2, \dots\}$ be the number of tosses required to get “head”. Compute the (unconditional) $\mathbb{E}(X)$ and $\text{Var}(X)$. (20p)
6. State and rigorously, from the axioms of probability, prove the “continuity of probability theorem”. (10p)
Let F_X be a cumulative distribution function of some random variable X . Using the above theorem, or otherwise, prove that the limit $\lim_{x \uparrow a} F(x)$ exists for each $a \in \mathbb{R}$. Does this limit *always* equal to $F(a)$?
If “yes” prove this fact, and if “not” specify the cases when the limit equals $F(a)$, and when it does not equal $F(a)$, respectively.

Good luck!

Solutions

- Note that since $A \in \mathcal{F}$ implies $A^c \in \mathcal{F}$, if $\mathcal{F}$ is finite it has even numbers of elements $(2, 4, \dots)$
 $\{\emptyset, \Omega\}; \{\{1, 2\}, \{3\}, \emptyset, \Omega\}; \{\{1, 3\}, \{2\}, \emptyset, \Omega\}; \{\{2, 3\}, \{1\}, \emptyset, \Omega\}$.
- Let D be the event that you have the disease, and T be the event that the test comes positive. We have $\mathbb{P}(T | D) = 0.99$, $\mathbb{P}(T | D^c) = 0.05$. By Bayes' theorem

$$\mathbb{P}(D | T) = \frac{\mathbb{P}(T | D) \mathbb{P}(D)}{\mathbb{P}(T)} = \frac{\mathbb{P}(T | D) \mathbb{P}(D)}{\mathbb{P}(T | D) \mathbb{P}(D) + \mathbb{P}(T | D^c) \mathbb{P}(D^c)}$$

Since $\mathbb{P}(D) = \alpha$, we have

$$\frac{1}{11} = \frac{0.99 \alpha}{0.99 \alpha + 0.05 (1 - \alpha)}$$

so $\alpha = \frac{1}{199}$.

- First of all, note $0 \leq Z \leq 2$. Let us compute the cdf of Z : $\mathbb{P}(Z \leq a) = \iint_{x+y \leq a, x+y \leq 1} dx dy$. For $0 \leq a \leq 1$ this is the area of the triangle with vertices $(0, 0)$, $(0, a)$ and $(a, 0)$ hence it's $a^2/2$. For $a \in [1, 2]$, one minus this integral is the area of the triangle with vertices $(1, 1)$, $(1, a-1)$, $(a-1, 1)$, i.e. $(2-a)^2/2$. Hence the density of Z is given by $F'_Z(a) =$

$$\begin{cases} a, & \text{if } 0 \leq a \leq 1; \\ 2-a, & \text{if } 1 \leq a \leq 2; \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Since $\int e^{-u^2/2} du = \sqrt{2\pi}$, we have

$$\begin{aligned} \iint c e^{-\frac{x^2+xy+y^2}{2}} dx dy &= c \int e^{-\frac{3y^2}{8}} \left[\int e^{-\frac{(x+y/2)^2}{2}} dx \right] dy = c \int e^{-\frac{3y^2}{8}} \left[\int e^{-\frac{v^2}{2}} dv \right] dy \\ &= c \sqrt{2\pi} \int e^{-\frac{3y^2}{8}} dy = c \sqrt{2\pi} \frac{2}{\sqrt{3}} \int e^{-\frac{v^2}{2}} dv = c \frac{4\pi}{\sqrt{3}} \end{aligned}$$

by changing variables $x := v - y/2$ and later $y := \frac{2}{\sqrt{3}}u$, so $c = \frac{\sqrt{3}}{4\pi}$ since the integral must be 1.

(b) Marginal density of Y is given by

$$f_Y(y) = \int f_{X,Y}(x, y) dx = \int c e^{-\frac{x^2+xy+y^2}{2}} dx = c e^{-\frac{3y^2}{8}} \left[\int e^{-\frac{(x+y/2)^2}{2}} dx \right] = c \sqrt{2\pi} e^{-\frac{3y^2}{8}}$$

so the conditional density of $X|Y = y$ is

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{e^{-\frac{x^2+xy+y^2}{2} + \frac{3y^2}{8}}}{\sqrt{2\pi}} = \frac{e^{-\frac{(x+y/2)^2}{2}}}{\sqrt{2\pi}}$$

that is, $X | Y = y$ is Normal with mean $-y/2$ and Variance = 1. Hence $\mathbb{E}(X^2 | Y = y) = \text{Var}(X^2 | Y = y) + [\mathbb{E}(X | Y = y)]^2 = 1 + y^2/4$.

- Conditionally on $Y = y$, the r.v. X is geometric with parameter y , so $\mathbb{E}(X | Y = y) = \frac{1}{y}$ and $\text{Var}(X | Y = y) = \frac{1-y}{y^2}$ (e.g. by formulas from the course). So

$$\mathbb{E}X = \mathbb{E}[\mathbb{E}(X | Y)] = \mathbb{E} \frac{1}{Y} = \int_0^1 \frac{1}{y} \cdot 2y dy = 2;$$

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[\text{Var}(X|Y)] + \text{Var}(\mathbb{E}(X|Y)) = \mathbb{E} \left[\frac{1}{Y^2} - \frac{1}{Y} \right] + \text{Var} \left(\frac{1}{Y} \right) \\ &= \mathbb{E} \frac{2}{Y^2} - \mathbb{E} \frac{1}{Y} - \left(\mathbb{E} \frac{1}{Y} \right)^2 = +\infty \quad \text{since } \mathbb{E} \left[\frac{2}{Y^2} \right] = \int_0^1 \frac{2}{y^2} \cdot 2y dy = \infty. \end{aligned}$$

- For the proof, please see the lecture notes. $\lim_{x \uparrow a} F(x)$ exists as an increasing bounded by 1 sequence; it equals $F(a-0) = \mathbb{P}(X < a) \neq \mathbb{P}(X \leq a) = F(a)$ unless $\mathbb{P}(X = a) = 0$.